

THE $e_k \star e_r$ PIERI RULE FOR ALL k : PARABOLIC KERNEL, CHAIN FACTORIZATION, AND A ONE-INDEX q -SERIES IDENTITY

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ABSTRACT. For every $k \geq 1$, $r \geq 0$ and every number of variables $m \geq 0$, we prove

$$t^{-\binom{k}{2}} e_k(Y) \bullet e_r = \sum_{b=0}^k s^b F_{k-b}(t^{r-b}) e_b e_{r+k-b}, \quad s = q^{-1},$$

with the explicit polynomial F_n defined in (1) below. Via Hikita's Definition 3.4 and Lemma 3.3 (arXiv:2503.23597), the left side is $e_k \star e_r$. The cases $k = 1$ and $k = 2$ recover Hikita's Theorem 3.12 and our W_r . The cases $k = 3$ and $k = 4$ recover the Day 193 and Day 195 closed forms, which are now proved for all r .

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1. CONVENTIONS AND STATEMENT

Operators. The conventions are those of the W_r note.

$$T_i F = t s_i F + (t-1) X_{i+1} \frac{s_i F - F}{X_i - X_{i+1}}, \quad \pi F = X_1 F(X_2, \dots, X_m, sX_1),$$

$$Y_i = t^{m-i} T_{i-1} \cdots T_1 \pi T_{m-1}^{-1} \cdots T_i^{-1}.$$

Symmetric functions.

- $a_{ij} = (X_i - tX_j)/(X_i - X_j)$ and $\prod_A^\times = \prod_{i \in A, j \notin A} a_{ij}$.
- $E(z) = \prod_i (1 + X_i z) = \sum_n e_n z^n$.
- $Q(y) = \sum_n q_n y^n = E(-ty)/E(-y)$.
- $[n] = [n]_t$.

Coefficients. For $0 \leq j \leq n$ put

$$(1) \quad \alpha_j = \frac{\prod_{i=1}^j (s - t^i)}{(t; t)_j}, \quad \alpha_{-1} = 0, \quad c(n, j) = \frac{(s; t)_{n-j}}{(t; t)_{n-j}} (\alpha_j - s t^{n-j} \alpha_{j-1}), \quad F_n(w) = \sum_{j=0}^n c(n, j) w^j.$$

Set $c(n, j) = 0$ outside the range $0 \leq j \leq n$.

Theorem 1. For all $m \geq 0$, $k \geq 1$ and $r \geq 0$,

$$\sum_{r \geq 0} z^r t^{-\binom{k}{2}} e_k(Y) \bullet e_r = \sum_{j+b \leq k} s^b t^{-kj} c(k-b, j) e_b z^{b-k} E(t^j z).$$

Taking the coefficient of z^r gives

$$t^{-\binom{k}{2}} e_k(Y) \bullet e_r = \sum_{b=0}^k s^b F_{k-b}(t^{r-b}) e_b e_{r+k-b}.$$

2. HECKE FACTS

The T_i satisfy the braid relations and $(T_i - t)(T_i + 1) = 0$. Hence:

- T_w is well defined on reduced words;
- $T_u T_v = T_{uv}$ whenever $\ell(uv) = \ell(u) + \ell(v)$.

For $J \subseteq S$, let W^J be the set of minimal left coset representatives, so that $w = w^J w_J$ uniquely and length-additively.

(Chain). If $K \subseteq J$, then $W^J \times (W_J)^K \rightarrow W^K$, $(u, v) \mapsto uv$, is a length-additive bijection.

- If $s \in K$, then $\ell(uvs) = \ell(u) + \ell(vs) = \ell(u) + \ell(v) + 1$.
- Conversely, if $x \in W^K$ but $x_J s < x_J$ for some $s \in K$, then $\ell(xs) < \ell(x)$, which is a contradiction.

Spherical vectors. If $T_k H = tH$ for all $s_k \in J$, then $T_v H = t^{\ell(v)} H$ for every $v \in W_J$.

Shifting through π . $\pi T_k = T_{k+1} \pi$ for $1 \leq k \leq m-2$.

Minimal representatives. Let $J_n = \langle s_i : i \neq n \rangle$. For $D = \{d_1 < \dots < d_n\}$, the word

$$w(D) = (s_{d_1-1} \dots s_1)(s_{d_2-1} \dots s_2) \dots (s_{d_n-1} \dots s_n)$$

sends $l \mapsto d_l$. Its length is $\sum_l (d_l - l) = \ell(w^D)$. Hence it is a reduced word for the minimal representative w^D . We set $\sigma^{(k)} = \sum_{|D|=k} T_{w(D)}$.

3. THE PARABOLIC KERNEL (A_k)

Lemma 2 (Key Lemma). *Let $1 \leq c < d_1 < \dots < d_n \leq m$, and let H satisfy $T_k H = tH$ for all $k \neq n$. Then*

$$(T_c \dots T_{m-1})^{-1} T_{w(D)} H = t^{n-(m-c)} T_{w(D-1)} H.$$

Proof. Let $x = s_c \dots s_{m-1}$. It sends $m \mapsto c$ and $i \mapsto i+1$ for $c \leq i < m$. Let $y = w(D-1)$; then $y(m) = m$.

- *Length.* x is increasing on $[1, m-1]$, and $xy(m) = c$ is inverted with exactly the $m-c$ positions i with $y(i) \geq c$. So $\ell(xy) = \ell(y) + m - c$.
- *Coset.* $xy([n]) = x(D-1) = D$. Therefore $xy = w^D v$ with $v \in W_{J_n}$ and $\ell(v) = m - c - n$. Hence $T_x T_y H = T_{w^D} T_v H = t^{m-c-n} T_{w^D} H$. $\square$

Proposition 3 (A_k). *Let F be symmetric and $b_1 < \dots < b_k$. Then $Y_{b_1} \dots Y_{b_k} F = t^{\binom{k}{2}} T_{w(b)} \pi^k F$. Consequently $e_k(Y) F = t^{\binom{k}{2}} \sigma^{(k)} \pi^k F$.*

Proof. Induct on k . Put $H = \pi^n F$; it is symmetric in the head and in the tail. By the Key Lemma,

$$Y_c T_{w(D)} H = t^{m-c} T_{c-1} \dots T_1 \pi (T_c \dots T_{m-1})^{-1} T_{w(D)} H = t^n T_{c-1} \dots T_1 \pi T_{w(D-1)} H.$$

Shifting through π turns this into $t^n T_{w(\{c\} \cup D)} \pi^{n+1} F$. Finally, $\binom{n}{2} + n = \binom{n+1}{2}$. $\square$

4. THE k -SET KERNEL (K_k)

Let $\sigma_{[l,m]} = \sum_{a=l}^m T_{a-1} \dots T_l$ and $K_l = \langle s_{l+1}, \dots, s_{m-1} \rangle$.

(C1) Two factorizations. Applying (Chain) along $K_k \subset K_{k-1} \subset \dots \subset K_0$, and also along $K_k \subset J_k$ (where $(W_{J_k})^{K_k} = S_k$), gives

$$\sum_{u \in W^{K_k}} T_u = \sigma_{[1,m]} \dots \sigma_{[k,m]} = \sigma^{(k)} \sum_{v \in S_k} T_v.$$

Consequently, if G is symmetric in the head and in the tail, then $\sigma_{[1,m]} \dots \sigma_{[k,m]} G = [k]! \sigma^{(k)} G$.

(C2) Ordered formula. Suppose G is symmetric in the tail only. Iterating Day 205b Lemma 1, applied first in $X_2, \dots, X_m$ over $\mathbb{Q}(s, t)(X_1)$ and then at the top level, gives

$$\sigma_{[1,m]} \dots \sigma_{[k,m]} G = \sum_{(a_1, \dots, a_k) \text{ distinct}} G(X_{a_1}, \dots, X_{a_k}; \text{rest}) \prod_{l=1}^k \prod_{j \notin \{a_1, \dots, a_l\}} a_{a_l j}.$$

(C3) Poincaré identity. Taking $m = k$ and $G = 1$ in (C1) and (C2) gives $\sum_{w \in S_k} \prod_{l < l'} a_{w_l w_{l'}} = [k]!$.

Proposition 4 (K_k). *If G is symmetric in the head and in the tail, then $\sigma^{(k)}G = \sum_{|A|=k} G^{(A)} \prod_A^\times$.*

Proof. In (C2), group the ordered tuples by their underlying set A . The weight of each tuple factors as $\prod_A^\times \cdot \prod_{l < l'} a_{a_l a_{l'}}$. Summing over the orderings of A gives $[k]! \prod_A^\times$ by (C3). Now compare with (C1). $\square$

5. THE GENERATING FUNCTION (E_k)

Let $\varphi(u) = u(1 + suz)/(1 + uz)$. By (A_k) and (K_k) ,

$$\Gamma_k := \sum_r z^r t^{-\binom{k}{2}} e_k(Y) \bullet e_r = E(z) \sum_{|A|=k} \prod_A^\times \prod_{a \in A} \varphi(X_a).$$

This holds for all $m \geq 0$; both sides vanish when $m < k$. Let T_k denote the right side of Theorem 1.

Lemma 5 (Recursion). *For $k, m \geq 1$,*

$$[k] \Gamma_k(X) = \sum_i X_i (1 + sX_i z) \prod_{j \neq i} a_{ij} \Gamma_{k-1}(\hat{X}_i).$$

Proof. Use (C2) and (C3), and peel off the first index $a_1 = i$. Then use $\varphi(X_i)E(X; z) = X_i(1 + sX_i z)E(\hat{X}_i; z)$. $\square$

Lemma 6 (Step). *For every $m \geq 0$ and $k \geq 1$,*

$$\sum_i X_i (1 + sX_i z) \prod_{j \neq i} a_{ij} T_{k-1}(\hat{X}_i) = [k] T_k(X).$$

Proof. Let $\gamma = t^j z$. Expand

$$e_b(\hat{X}_i) = \sum_c (-X_i)^c e_{b-c}, \quad E(\hat{X}_i; \gamma) = \frac{E(\gamma)}{1 + \gamma X_i},$$

and split

$$\frac{1 + suz}{1 + \gamma u} = st^{-j} + \frac{1 - st^{-j}}{1 + \gamma u}.$$

Now apply Lemma 2 in the form $\sum_i g(X_i) \prod_{j \neq i} a_{ij} = (1 - t)^{-1} \Omega[g]$, where $\Omega[u^n] = q_n$ for $n \geq 1$. Two facts evaluate the result:

- (a) $\sum_c (-1)^c e_{b-c} q_{c+1} = (1 - t^{b+1}) e_{b+1}$;
- (b) $E(\gamma) \sum_c (-1)^c e_{b-c} \Omega\left[\frac{u^{1+c}}{1 + \gamma u}\right] = [w^b] \frac{E(w)E(t\gamma) - E(\gamma)E(tw)}{w - \gamma}$
 $= - \sum_{n \leq b} e_n \gamma^{n-1-b} (E(t\gamma) - t^n E(\gamma)).$

For (b), first note that $\Omega[u/((1+wu)(1+\gamma u))] = (Q(-\gamma) - Q(-w))/(w - \gamma)$; then use $E(x)Q(-x) = E(tx)$.

Write $N'_{jb} = s^b t^{-(k-1)j} c(k-1-b, j)$ for the coefficients of T_{k-1} , and put $n = k - b'$. The coefficient of $e_{b'} z^{b'-k} E(t^j z)$ on the left side is then $(1 - t)^{-1} s^{b'} t^{-kj}$ times

$$(1 - t^{b'}) c(n, j) + t^{b'} \left[(1 - st^{-j}) \sum_{p \geq 0} (st^{-j})^p c(n-1-p, j) - t^n (1 - st^{1-j}) \sum_{p \geq 0} (st^{1-j})^p c(n-1-p, j-1) \right].$$

By Lemma 7 the bracket equals $(1 - t^n) c(n, j)$. So the whole expression is $(1 - t^k) c(n, j)$, which is the claim. $\square$

Lemma 7 ($\star$). For $n \geq 1$ and $j \geq 0$,

$$(1 - t^n)c(n, j) = (1 - st^{-j}) \sum_{p \geq 0} (st^{-j})^p c(n - 1 - p, j) - t^n (1 - st^{1-j}) \sum_{p \geq 0} (st^{1-j})^p c(n - 1 - p, j - 1).$$

Proof. Generating functions. Let $G(x) = \sum_n \frac{(s;t)_n}{(t;t)_n} x^n$. Comparing coefficients gives $(1 - x)G(x) = (1 - sx)G(tx)$. Let $C_j(x) = \sum_n c(n, j)x^n$. Then

$$C_j(x) = x^j [\alpha_j G(x) - s\alpha_{j-1} G(tx)].$$

Reformulation. In terms of the C_j , the claim is

$$\frac{C_j(x)(1 - x)}{1 - st^{-j}x} - C_j(tx) = -\frac{(1 - st^{1-j})tx C_{j-1}(tx)}{1 - st^{2-j}x}.$$

Linear factorization. Put $D_j = \alpha_j - s\alpha_{j-1}$. Since $\alpha_j = \alpha_{j-1}(s - t^j)/(1 - t^j)$, we have

$$\alpha_j(1 - sy) - s\alpha_{j-1}(1 - y) = D_j(1 - st^{-j}y).$$

Indeed, both sides are linear in y and agree at $y = 0$ and at $y = t^j/s$.

Reduction. Substituting the factorization, both sides of the reformulated claim become the common factor $x^j G(t^2x)/(1 - tx)$ times

$$D_j(1 - t^j) \quad \text{and} \quad t(s - t^{j-1})D_{j-1}$$

respectively. These are equal, because $D_j = \alpha_{j-1}t^j(s - 1)/(1 - t^j)$ for $j \geq 1$. For $j = 0$ both sides vanish. $\square$

Proof of Theorem 1. Induct on k , for all m simultaneously. The base case is $\Gamma_0 = E(z) = T_0$. For $k \geq 1$:

- if $m = 0$, the Step Lemma gives $T_k = 0 = \Gamma_k$;
- if $m \geq 1$, combine the Recursion, the induction hypothesis in $m - 1$ variables, and the Step Lemma.

Finally, $[z^r] z^{b-k} E(t^j z) = t^{j(r+k-b)} e_{r+k-b}$. $\square$

6. REMARKS

What is classical and what is new. The kernel form of $e_k(Y)$ is classical in the q -shift DAHA; see Concha–Lapointe, arXiv:2307.02385, Lemmas 8 and 10. The level-one π^k transfer and the e -basis Pieri rule for general k are, as far as the Day 207 audit found, new.

Specializations. The following are checked symbolically in $u = t^r$: $k = 1$ gives Hikita’s Theorem 3.12; $k = 2$ gives W_r ; $k = 3, 4$ give Days 193 and 195. The case $r = 0$ recovers Hikita’s Lemma 3.3, $e_k(Y) \bullet 1 = t^{\binom{k}{2}} e_k$; this is checked symbolically for $k \leq 6$.

Machine checks. (`scripts/day207b/`, all OK).

- Key Lemma and (A_k) : exact for $m \leq 6$.
- (C2): checked.
- Lemma $\star$: checked for $n \leq 9$.
- Step Lemma: checked with z symbolic.
- Full theorem against a direct AHA computation: 216 cases, including $m < k$.

Open. The ${}_2\phi_1$ form $F_n(w) = (1 - t^n w) \frac{(s;t)_n}{(t;t)_n} {}_2\phi_1(t^{1-n}, t/s; t^{1-n}/s; t; w)$ is only computed ($n \leq 6$), and it is not needed for the theorem.